

★ WEEK 13 · PART 3 · PRACTICE

Credible Intervals and Prediction, by Hand

A Direct Probability, and an Honest Forecast

The credible interval and the posterior predictive

Part 3's Bayesian outputs, with the arithmetic shown. We compute a 95% credible interval on an expected return - a direct probability statement - then build a posterior predictive interval that is honestly wider because it carries the estimation uncertainty.

3P

PART 3 · APPLICATIONS

From Posterior to Decision

Case A: a 95% credible interval on an expected return, and how it differs from a confidence interval. Case B: the posterior predictive - a forecast interval that adds parameter uncertainty and is honestly wider than the plug-in. Then a Claude Code lab.

What Part 3 Told Us

Now we compute the outputs - with numbers

1

A credible interval is a direct probability

It says the parameter is in this range with 95% probability, given the data - the statement people actually want.

2

The posterior predictive forecasts honestly

It forecasts the next outcome by integrating over the posterior, so it carries both outcome and estimation uncertainty.

3

Bayes suits finance, at a cost

Direct probabilities and honest forecasts are ideal for decisions - but exact computation doesn't scale, which is why MCMC exists.

Everything here is a calculator exercise: one posterior, one credible interval, one forecast.

How These Cases Work

Data, then arithmetic, then meaning

The shared setup

- Posterior on the mean return: $N(10\%, 8\%)$.
- Year-to-year return volatility: 15%.
- A short history, so real uncertainty.

What you should watch

- How the credible interval reads directly.
- How the predictive adds parameter uncertainty.
- Why the predictive interval is wider.

The Credible Interval

A probability about the parameter

Our posterior on a strategy's expected return is centered at 10% with an 8% spread. From it, a 95% credible interval is a direct statement: there is a 95% probability the true expected return lies in this range, given the data. That's exactly what a decision-maker wants - and it's not what a confidence interval says

The posterior

- Expected return centered at 10%.
- Posterior spread of 8%.
- Reflecting a short track record.

What we'll do by hand

- Take the central 95% of the posterior.
- Read off the interval.
- Contrast it with a confidence interval.

The Credible Interval

Direct probability about the parameter

95% credible interval : $P(a \leq \theta \leq b \mid \text{data}) = 0.95$ (a direct probability about the parameter)

How to read it

- It's a probability about the parameter itself.
- Conditioned on the data you actually saw.
- The natural interval for a decision.

How to build it

- Take the central 95% of the posterior.
- For a Gaussian, mean plus or minus 1.96 sd.
- The endpoints are the interval.

The 95% Credible Interval

Central band of the posterior

$$[a, b] = \mu_{\text{post}} \pm 1.96 \sigma_{\text{post}} = 10\% \pm 1.96(8\%) = [-5.7\%, 25.7\%]$$

The calculation

- 1.96 times the 8% spread is 15.7%.
- Add and subtract from the 10% center.
- The interval is -5.7% to 25.7%.

The reading

- A 95% probability the mean is in this band.
- It's wide - the history is short.
- The band even includes negative returns.

Credible vs Confidence

Two intervals, two meanings

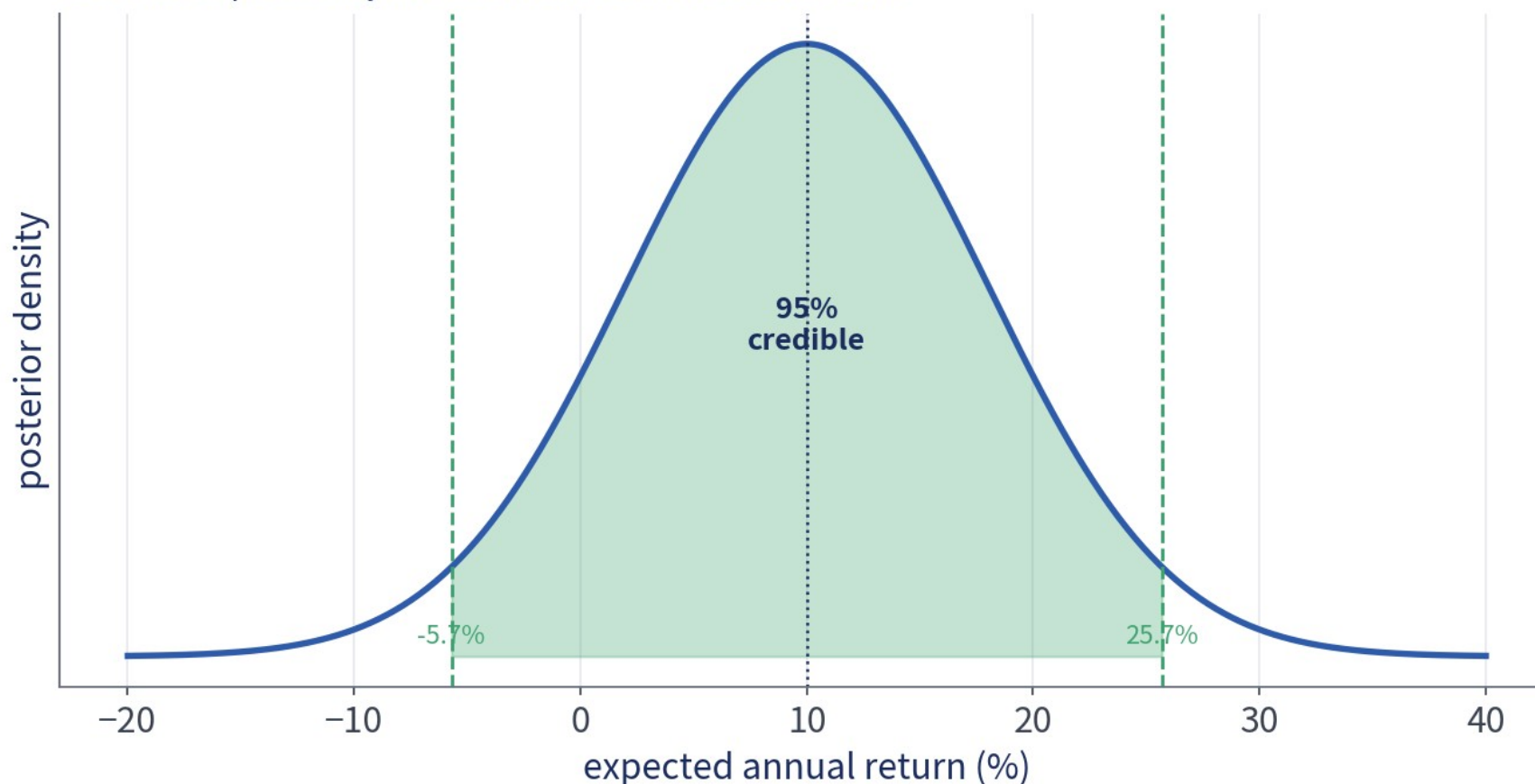
credible: 95% probability $\theta \in [a, b]$ given this data
confidence: 95% of such intervals cover θ (a statement about the procedure)

The credible interval makes a probability statement about the parameter given this data. The confidence interval instead describes the long-run behavior of the procedure - not this specific interval.

The Central 95%

The shaded band holds the central 95% of the posterior. Its interpretation is direct: given the data, there's a 95% probability the true expected return falls between -5.7% and 25.7%. The width reflects genuine uncertainty from the short history.

Case A: 95% probability the true mean lies in the shaded band



Why the Distinction Matters

Say what you mean about probability

What people want

- The chance the parameter is in a range.
- A statement about this data, this estimate.
- That's the credible interval.

What confidence gives

- A guarantee about repeated procedures.
- Not a probability about this interval.
- Useful, but a different question.

The Posterior Predictive

Forecasting with honest uncertainty

To forecast next year's return, a naive approach plugs in the 10% estimate and uses the 15% return volatility. But that ignores the uncertainty in the estimate itself. The posterior predictive adds it in: the forecast variance is the outcome variance plus the parameter variance - so the interval is honestly wider. We compute both

The two uncertainties

- Outcome noise: returns vary 15% year to year.
- Parameter uncertainty: the mean's 8% spread.
- Both belong in an honest forecast.

What we'll do by hand

- Add the two variances.
- Take the square root for the predictive sd.
- Compare the interval to the plug-in.

The Posterior Predictive

Outcome noise plus parameter uncertainty

$$\text{Var}(\text{next return}) = \underbrace{\sigma_{\text{data}}^2}_{\text{outcome noise}} + \underbrace{\sigma_{\text{post}}^2}_{\text{parameter uncertainty}}$$

The plug-in error

- Using only the outcome volatility.
- It pretends the mean is known exactly.
- So its interval is too narrow.

The predictive fix

- Integrate the forecast over the posterior.
- That adds the parameter variance.
- The total is the predictive variance.

The Predictive SD

Combine the two spreads

$$\sigma_{\text{pred}} = \sqrt{\sigma_{\text{data}}^2 + \sigma_{\text{post}}^2} = \sqrt{15^2 + 8^2} = \sqrt{289} = \mathbf{17\%}$$

Adding the variances - 15-squared plus 8-squared - gives 289, whose square root is exactly 17%. The predictive spread is bigger than the 15% outcome volatility alone.

Predictive vs Plug-In

How much wider the honest interval is

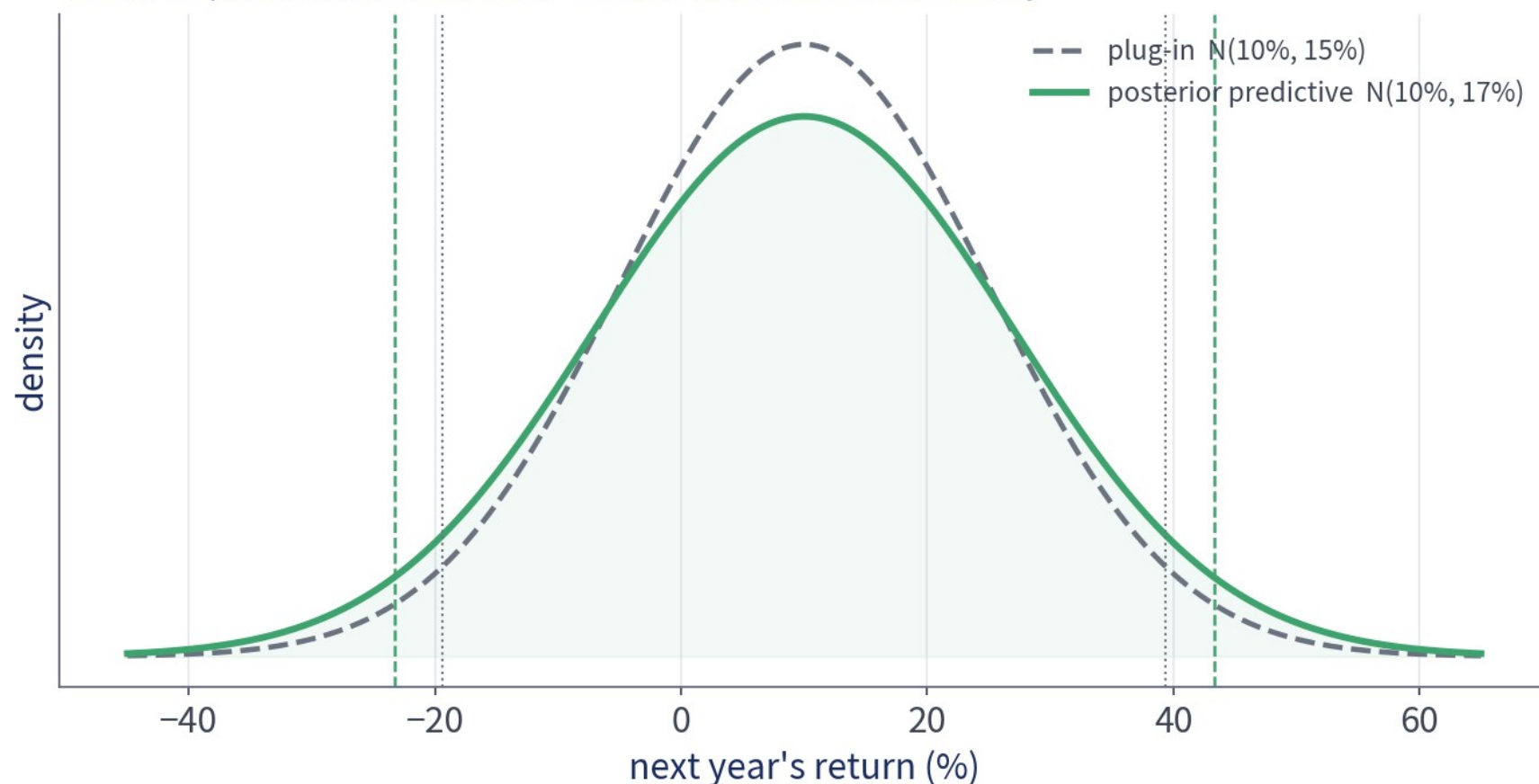
$$\begin{aligned}\text{plug-in: } & 10\% \pm 1.96(15\%) = [-19.4\%, 39.4\%] \\ \text{predictive: } & 10\% \pm 1.96(17\%) = [-\mathbf{23.3\%}, \mathbf{43.3\%}] \text{ (wider)}\end{aligned}$$

The plug-in forecast interval runs -19.4% to 39.4%; the posterior predictive runs -23.3% to 43.3% - wider on both ends, because it admits we don't know the mean exactly.

Honestly Wider

Both forecasts center on 10%, but the posterior predictive (green) has fatter tails than the plug-in (dashed) - carrying the extra uncertainty about the mean. Ignoring that estimation error would leave you overconfident about next year's range of outcomes

Case B: the predictive interval is wider - it carries the estimation uncertainty



Why Honest Intervals Matter

Overconfidence is expensive

The plug-in trap

- Treating the estimated mean as the truth.
- Reporting a forecast interval too narrow.
- Underpricing tail risk.

The predictive fix

- Carry the estimation uncertainty forward.
- Widen the interval accordingly.
- Especially important on short histories.

Why Bayes Suits Finance

And the catch that shapes its history

Both outputs are what finance actually needs: a direct probability about a parameter, and a forecast that owns its uncertainty. That's why Bayesian methods fit decision-making so well. The catch is computation - exact posteriors don't scale beyond simple models, which is why the modern toolkit leans on MCMC to sample what it

in

The strengths

- Credible intervals: direct probability.
- Posterior predictive: honest forecasts.
- Natural for sizing and risk.

The catch

- Exact Bayes doesn't scale with dimensions.
- The evidence integral becomes intractable.
- MCMC samples the posterior instead.

Reproduce the Interval and the Forecast

Direct Claude Code to check your arithmetic

GOAL Verify the credible interval and the posterior predictive, then compare to the plug-in.

[INPUT] Prompt · Claude Code

```
Write and run part3_practice.py that:  
CASE A - credible interval:  
* from posterior  $N(0.10, 0.08)$ , print the 95%  
  credible interval (mean  $\pm 1.96$  sd).  
CASE B - posterior predictive:  
* with outcome vol 0.15 and mean uncertainty 0.08,  
  print the predictive sd =  $\sqrt{0.15^2 + 0.08^2}$ ;  
* print the plug-in and predictive 95% intervals.  
Then plot the posterior with its credible band, and  
the plug-in vs predictive densities.  
Match every printed number to the slides.
```

Check against the slides

- credible: [-5.7%, 25.7%].
- predictive sd = 17%.
- plug-in: [-19.4%, 39.4%].
- predictive: [-23.3%, 43.3%].

Deliverable: part3_practice.py whose printed numbers match the credible interval and the forecasts.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 What does a credible interval say?

95% probability the parameter is in the range

Q2 Why is the predictive wider than the plug-in?

It adds the parameter uncertainty

Q3 Why does exact Bayes need MCMC?

The posterior doesn't scale with dimensions

All three come from the credible interval and the posterior predictive.

Three Pitfalls to Avoid

Where the outputs trip people

Common mistakes

- "A confidence interval gives the probability here."
- "Plug in the estimate and use outcome vol."
- "Exact Bayes works for any model."

The reality

- Only the credible interval is a direct probability.
- The predictive adds parameter uncertainty - it's wider.
- Beyond simple models, you need MCMC to sample it.

Two Outputs to Remember

Part 3 applications in one screen

1

The credible interval is a direct probability

From a posterior of $N(10\%, 8\%)$, the 95% credible interval is -5.7% to 25.7% - a direct statement that the true mean is in that band with 95% probability, given the data. A confidence interval, by contrast, describes only the procedure.

2

The posterior predictive is honestly wider

Forecasting next year's return, the predictive sd combines the 15% outcome volatility with the 8% parameter uncertainty into 17%, widening the interval from -19.4%-39.4% (plug-in) to -23.3%-43.3%. Ignoring estimation error breeds overconfidence.

3

Bayes fits finance, MCMC makes it scale

Direct probabilities and honest forecasts are exactly what decisions need. But exact posteriors don't scale, so the modern toolkit samples them with MCMC - the natural next step.

Next — Week 14: sampling the posterior when it has no closed form, with Markov chain Monte Carlo.